

ROS2 Gazebo Robot Modeling Guide (Draft)

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1 Introduction

This guide is a draft of the ROS2 Gazebo Robot Modeling Guide. It is a work in progress and is not yet complete.

2 Setup

- Make sure you have a Ubuntu/Debian system:
 - For windows users (Not tested yet), you can use the Windows Subsystem for Linux (WSL) to run a Ubuntu/Debian system.

```
wsl --install -d Ubuntu-22.04
wsl --set-version Ubuntu-22.04 2
wsl --set-default-version 2
wsl --set-default Ubuntu-22.04
```

- For MacOS users (Missing additional steps), you can use Parallel Desktop to run a Ubuntu/Debian system.
- Make sure you have VSCode installed. Also ensure that VSCode has the `code` shell command enabled. Please refer to the VSCode documentation: [VSCode Documentation](#)
- Make sure you have Docker Daemon installed.
 - In Ubuntu/Debian systems, you can install it via script provided by Docker convenient script [Docker Installation Guide](#).

```
curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh ./get-docker.sh --dry-run
```

- For windows users (Not tested yet), you can install Docker Desktop from the Microsoft Store.
- Follow post installation steps to remove the need for sudo to run Docker commands: [Docker Post-Installation Steps](#).
- Clone the repository: [ROS2_Docker_UI](#):

```
git clone https://github.com/aldaajo92/ROS2\_Docker\_UI.git
cd ROS2\_Docker\_UI
git checkout unal
code .
```

- Follow the instructions in the README.md file to build and run the Docker container:

```
./scripts/build.sh
./scripts/run.sh # Run the main container and run the main bash
                  shell
```

If you need to open a new terminal for the container, you can use the following command:

```
./scripts/bash.sh # In a new terminal, run the bash shell
```

It is important that you identify when you are in the container. For example, if you execute the `./scripts/run.sh` command, you should see the following in your terminal:

```
$ ./scripts/run.sh
Welcome to ROS2 docker container
dockeruser@HostComputer:~/ros2_ws$
```

It means that you are in the container and located in the `ros2_ws` folder. The container acts as another computer, so you can execute commands on it. The `ros2_ws` folder is located in the path `/home/dockeruser/ros2_ws` in the container. This folder is shared with the host computer, so you can interact with this folder using an editor like VSCode.

3 ROS2 Workspace

The default ROS2 workspace is located in the `ros2_ws` folder. This folder has the following structure:

```
ros2_ws/
|-- build/
|-- install/
|-- src/
```

Where:

- `build/`: Contains the build files for the workspace.
- `install/`: Contains the installed files for the workspace.
- `src/`: Contains the ros2 packages for the workspace.

The `ros2_ws/src` folder should contain the ros2 packages, to create a new package, you can use the following command:

```
ros2 pkg create --build-type ament_cmake <package_name>
```

It is common to pass the dependencies on the package creation command, for example:

```
ros2 pkg create --build-type ament_cmake --dependencies rclcpp std_msgs
              geometry_msgs
```

4 Basic ROS2 Commands

The following are some basic ROS2 commands to get you started with the `ROS2_Docker_UI` container.

4.1 Source and Build

The source commands means to load the ROS2 environment variables into the current shell.

- **Source the ROS2 environment before performing any ROS2 commands:**

```
source /opt/ros/humble/setup.bash
```

- **Build the default ros2_ws workspace:**

```
source /opt/ros/humble/setup.bash
colcon build
source install/setup.bash
```

- **Build a single package:**

```
colcon build --packages-select <package_name>
source install/setup.bash
```

Please note that a new source command is needed after a build command.

4.2 Run Nodes and Launch Files

The following are some basic ROS2 commands to run nodes and launch files.

- **Run a node:**

```
ros2 run <package_name> <node_name>
```

- **Run a launch file:**

```
ros2 launch <package_name> <launch_file_name>
```

4.3 Exercise

- clone the ROS2 package in the ros2_ws/src folder:

```
git clone https://github.com/alda92/waver_description_ros2
waver_description
```

- build the package:

```
colcon build --packages-select waver_description
```